

Assignment No. 19

FVM practice

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Solution to 4.2

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Assignment 19.2 Vensteeg, Malalasekara

Example 4.2 Now we discuss a problem that includes source other than those arising from boundary conditions. Figure 4.6 shows

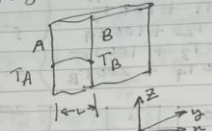


Fig. 4.6

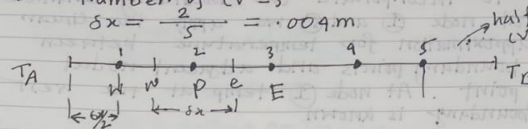
a large plate of thickness $L = 2\text{ cm}$ with constant thermal conductivity $k = 0.5\text{ W/mK}$ and uniform heat generation $q = 1000\text{ kW/m}^3$. The faces A and B are at temperatures 50°C and 250°C respectively. Assuming the dimensions in y and z directions are so large that temperature gradients are significant in the x direction only calculate the steady state temperature distribution compare the numerical result with analytical solution. The governing equation

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) + q = 0$$

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solⁿ: Number of CV = 5

$$\delta x = \frac{2}{5} = 0.004 \text{ m}$$



$$\int_{\Delta V} \frac{d}{dx} \left(k A \frac{dT}{dx} \right) dV + \int_{\Delta V} q dV = 0$$

$$\left(k A \frac{dT}{dx} \right)_e - \left(k A \frac{dT}{dx} \right)_w + q \Delta V = 0$$

$$\Rightarrow \left[k_e A \left(\frac{T_E - T_P}{\delta x} \right) - k_w A \left(\frac{T_P - T_W}{\delta x} \right) \right] + q A \delta x = 0$$

This can be arranged as,

$$\left(\frac{k_e A}{\delta x} + \frac{k_w A}{\delta x} \right) T_P = \left(\frac{k_w A}{\delta x} \right) T_W + \left(\frac{k_e A}{\delta x} \right) T_E + q A \delta x$$

$$\Rightarrow a_P T_P = a_W T_W + a_E T_E + S_u$$

As, $k_e = k_w = k$ we have

a_W	a_E	a_P	S_P	S_u
$\frac{kA}{\delta x}$	$\frac{kA}{\delta x}$	$a_W + a_E - S_P$	0	$qA\delta x$

→ valid for node (2), (3), (4)



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To incorporate the boundary conditions at node ① and ③ we apply the linear approximation for temperature between boundary points and adjacent nodal point. At node ① temp at the west boundary is known

For node ① $\Rightarrow$

$$\left[\left(kA \frac{dT}{dx} \right)_e - \left(kA \frac{dT}{dx} \right)_w \right] + qA\Delta x = 0$$

Linear approximation for temp between A and P yields

$$\left\{ \left[k_e A \left(\frac{T_E - T_P}{\delta x} \right) \right] - \left[k_A A \left(\frac{T_P - T_A}{\delta x} \right) \right] \right\} + qA\delta x = 0$$

The above equation can be arranged, using $k_e = k_A = k$, to yield discretized equation for boundary node 1

$$a_P T_P = a_W T_W + a_E T_E + S_u$$

a_W	a_E	a_P	S_P	S_u
0	$\frac{kA}{\delta x}$	$a_W + a_E = 2P$	$-\frac{2kA}{\delta x}$	$qA\delta x + \frac{2kA}{\delta x} T_A$



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similarly for Node-5

$$\left[k_B A \left(\frac{T_B - T_P}{\delta x / 2} \right) - k_W A \left(\frac{T_P - T_W}{\delta x} \right) \right] + q A \delta x = 0$$

$k_B = k_W = k$

$$a_P T_P = a_W T_W + a_E T_E + S_u$$

a_W $\frac{kA}{\delta x}$	a_E	a_P	S_P	S_u
	0	$a_W + a_E$ $-S_P$	$\frac{-2kA}{\delta x}$	$qA\delta x + \frac{2kA}{\delta x} T_B$

$A = \text{unit area}$
 $k = 0.5 \text{ W/m}\cdot\text{K}$
 $q = 1000 \text{ kW/m}^3$
 $\delta x = 0.04$

Node	a_W	a_E	S_u	S_P	$a_P = a_W + a_E - S_P$
1	0	125	$\frac{4000}{\delta x} + \frac{250}{T_A}$	-250	375
2	125	125	4000	0	250
3	125	125	4000	0	250
4	125	125	4000	0	250
5	125	0	$\frac{4000}{\delta x} + \frac{250}{T_B}$	-250	375

Now,
for Node 1

$$375 T_1 = 125 T_2 + 16500 \quad \text{--- (i)}$$

for Node 2, $-125 T_1 + 250 T_2 - 125 T_3 = 9000 \quad \text{--- (ii)}$

Node 3 $-125 T_2 + 250 T_3 - 125 T_4 = 9000 \quad \text{--- (iii)}$

Node 4 $-125 T_3 + 250 T_4 - 125 T_5 = 9000 \quad \text{--- (iv)}$

Node 5, $375 T_5 = 125 T_4 + 66500 \quad \text{--- (v)}$

Augmented Matrix

$$\begin{bmatrix} 375 & -125 & 0 & 0 & 0 & : & 16500 \\ -125 & 250 & -125 & 0 & 0 & : & 9000 \\ 0 & -125 & 250 & -125 & 0 & : & 9000 \\ 0 & 0 & -125 & 250 & -125 & : & 9000 \\ 0 & 0 & 0 & -125 & 375 & : & 66500 \end{bmatrix}$$

Applying GA method (module prepared earlier)

$$\begin{aligned} T_1 &= 110 \\ T_2 &= 198 \\ T_3 &= 284 \\ T_4 &= 278 \\ T_5 &= 270 \end{aligned}$$

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We get the Analytical solution by integrating the GTE twice

$$T = \left[\frac{T_B - T_A}{L} + \frac{q_{pk}}{2k} (L - x) \right] x + T_A$$